Practice exam: sample solution

Microeconomics I · Part I · 100 points: Part A 40, Part B 30, Part C 30

In Parts B and C a full answer shows the working that leads to the result, as below; in the exam an answer that your calculation does not support receives only partial points or none. Any correct method is fine, not only the one shown. As on page 2 of the paper, CV is the compensating variation and EV the equivalent variation of a price change.

Part A (multiple choice, 40 points)

Key: A1 C, A2 A, A3 A, A4 B, A5 B, A6 D, A7 A, A8 C, A9 C, A10 B, A11 D, A12 D.

A1 [4 points] **Answer C:** Complete: no. Transitive: no.

Why. Not complete: completeness with $y = x$ needs $x \succsim x$, that is $2x_1 + x_2 \geq x_1 + 2x_2$, i.e. $x_1 \geq x_2$; for $x = (0, 1)$ this is $1 \geq 2$, false (complete $\Rightarrow$ reflexive). Not transitive: $x = (0, 2)$, $y = (2, 0)$, $z = (4, 0)$: $x \succsim y$ ($2 \geq 2$) and $y \succsim z$ ($4 \geq 4$), but $x \succsim z$ would need $2 \geq 4$.

Other options: A, transitivity fails: $(0, 2) \succsim (2, 0) \succsim (4, 0)$, but not $(0, 2) \succsim (4, 0)$; B, both fail; D, completeness fails: it needs $x \succsim x$ for every x, and $x = (0, 1)$ gives $1 \geq 2$.

A2 [3 points] **Answer A:** They are convex, but not strictly convex.

Why. u is a minimum of linear functions, hence concave, hence quasiconcave: convex preferences. The indifference curves consist of straight pieces, so not strictly convex. At $(2, 1)$, $u = \min\{5, 3\} = 3$ and $(3, 1)$ also gives 3: not strongly monotone.

Other options: B, kinks do not break convexity: a minimum of linear functions is concave, so upper contour sets are convex; C, at $(2, 1)$ more good 1 alone does not raise u; D, straight pieces of the indifference curves.

A3 [3 points] **Answer A:** $\frac{1}{2}$

Why. The marginal rate of substitution is $\mathrm{MRS}_{12} = \dfrac{\frac{1}{2}x_1^{-1/2}x_2^{3/2}}{\frac{3}{2}x_1^{1/2}x_2^{1/2}} = \dfrac{x_2}{3x_1} = \dfrac{3}{6} = \frac{1}{2}$.

Other options: B, multiplies by the exponent ratio instead of dividing: $3x_2/x_1$; C, inverted ratio; D, drops the exponents: x_2/x_1.

A4 [3 points] **Answer B:** $\ln u(x)$

Why. Any strictly increasing f keeps the representation. $f(t) = \ln t$ has $f'(t) = 1/t > 0$. The others are not increasing on $(0, 1)$.

Other options: A, falls then rises on $(0, 1)$: not increasing; C, decreasing: reverses the ranking; D, rises then falls on $(0, 1)$: not increasing.

A5 [3 points] **Answer B:** $(0,\ 9)$

Why. Perfect substitutes: compare utils per euro, $4/3$ for good 1 and $3/2$ for good 2. All wealth goes to good 2: $(0,\ 18/2) = (0, 9)$.

Other options: A, compares $4 > 3$ without dividing by the prices; C, a point on the budget line in the ratio 4 : 3 of the weights; D, splits her wealth equally.

A6 [4 points] **Answer D:** $\partial x_2/\partial p_1 = \frac{7}{2}$ and $\partial x_2/\partial w = \frac{1}{8}$

Why. Local non-satiation gives Walras' law $p_1x_1(p, w) + p_2x_2(p, w) = w$ at every (p, w). Differentiate in p_1 (Cournot aggregation): $p_1\,\partial x_1/\partial p_1 + p_2\,\partial x_2/\partial p_1 + x_1 = 0$, so $9 \cdot (-1) + 2\,\partial x_2/\partial p_1 + 2 = 0$ and $\partial x_2/\partial p_1 = \frac{7}{2}$. Differentiate in w (Engel aggregation): $p_1\,\partial x_1/\partial w + p_2\,\partial x_2/\partial w = 1$, so $9 \cdot \frac{1}{12} + 2\,\partial x_2/\partial w = 1$ and $\partial x_2/\partial w = \frac{1}{8}$.

Other options: A, Cournot aggregation without the term x_1: $9 \cdot (-1) + 2\,\partial x_2/\partial p_1 = 0$; B, both mistakes; C, Engel aggregation without the prices: $\partial x_1/\partial w + \partial x_2/\partial w = 1$.

A7 [3 points] **Answer A:** It is homogeneous of degree 1 in p.

Why. Mas-Colell, Whinston and Green (MWG), Proposition 3.E.2: e is homogeneous of degree 1 and concave in p, strictly increasing in u.

Other options: B, that is Hicksian demand $h(p, u)$; C, it is concave in p; D, it is strictly increasing in u.

A8 [4 points] **Answer C:** $CV < EV < 0$

Why. Both are negative after a price rise. $-CV$ is the area left of $h_1(\cdot, u^0)$ and $-EV$ the area left of $h_1(\cdot, u^1)$; with a normal good $h_1(\cdot, u^0)$ lies to the right, so $CV < EV < 0$.

Other options: A, the order for an inferior good; B, a price rise is a loss: both negative; D, only without a wealth effect (quasilinear).

A9 [3 points] **Answer C:** $c(F,u) < E[x]$

Why. Jensen with strict concavity: $u(c(F,u)) = E[u(x)] < u(E[x])$, and u is increasing, so $c(F,u) < E[x]$.

Other options: A, risk-loving; B, $u(c)$ equals $E[u(x)]$ by definition; D, risk-loving.

A10 [3 points] **Answer B:** Continuity.

Why. Take $L = 100$ euros for sure, $L' = 50$ euros for sure and $L'' = 0$ euros for sure, so $L \succ L' \succ L''$. Every mix $\alpha L + (1-\alpha)L''$ with $\alpha < 1$ has a positive chance of 0 euros, so L' is better; only $\alpha = 1$ beats L'. The set $\{\alpha : L' \succsim \alpha L + (1-\alpha)L''\} = [0,1)$ is not closed: continuity fails (MWG's safety-first example). Both criteria are linear in probabilities, so independence holds.

Other options: A, a lexicographic order is transitive; C, mixing with a common lottery scales both probabilities equally, so rankings stay; D, he ranks every pair.

A11 [4 points] **Answer D:** $(\frac{1}{4},\ \frac{3}{4},\ 0)$

Why. The indifference gives $u(90) = \frac{4}{5}u(240) + \frac{1}{5}u(0) = \frac{4}{5}$, so a lottery (p_1, p_2, p_3) has expected utility $p_1 + \frac{4}{5}p_2$: $(\frac{1}{4}, \frac{3}{4}, 0)$ gives $\frac{1}{4} + \frac{3}{5} = \frac{17}{20} = 0.85$; $(\frac{3}{4}, 0, \frac{1}{4})$ gives 0.75; $(\frac{1}{2}, \frac{1}{5}, \frac{3}{10})$ gives $0.5 + 0.16 = 0.66$; $(0, \frac{4}{5}, \frac{1}{5})$ gives 0.64.

Other options: A, expected utility $\frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$; B, expected utility $\frac{3}{4} = \frac{15}{20}$: the highest expected prize (180 euros), not the highest expected utility; C, expected utility $\frac{1}{2} + \frac{1}{5} \cdot \frac{4}{5} = \frac{33}{50}$.

A12 [3 points] **Answer D:** G first-order stochastically dominates F.

Why. Cumulative distribution functions (CDFs): $F = \frac{1}{2}$ on $[5,45)$; $G = \frac{1}{4}$ on $[5,25)$ and $\frac{1}{2}$ on $[25,45)$. So $G(x) \le F(x)$ everywhere: G dominates F.

Other options: A, they touch but never cross; B, means are 25 and 30; C, the higher CDF is the dominated one.

Part B (numerical answers, 30 points)

B1 [4 points] Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1,x_2) = \sqrt{x_1} + 3\sqrt{x_2}$. Find x_1 such that the bundle $(x_1, 1)$ is indifferent to the bundle $(16, 4)$.

Answer: $x_1 = 49$

Working. $u(16,4) = 4 + 6 = 10$; $\sqrt{x_1} + 3 = 10$ gives $\sqrt{x_1} = 7$, $x_1 = 49$.

B2 [4 points] Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1,x_2) = x_1^{1/3} + x_2^{1/3}$. The marginal rate of substitution is $\mathrm{MRS}_{12}(x) = \dfrac{\partial u/\partial x_1}{\partial u/\partial x_2}$. What is MRS_{12} at the bundle $x = (8, 27)$?

Answer: $\mathrm{MRS}_{12}(8,27) = \frac{9}{4}$

Working. $\partial u/\partial x_1 = \frac{1}{3}x_1^{-2/3}$ and $\partial u/\partial x_2 = \frac{1}{3}x_2^{-2/3}$, so $\mathrm{MRS}_{12} = (x_2/x_1)^{2/3} = (27/8)^{2/3} = (3/2)^2 = \frac{9}{4}$ ($8^{1/3} = 2$, $27^{1/3} = 3$).

B3 [4 points] A consumer has utility $u(x_1,x_2) = 3\ln x_1 + 2\ln x_2$, prices $p = (3,4)$ and wealth $w = 25$. How many units of good 1 does she buy?

Answer: $x_1 = 5$

Working. Cobb–Douglas shares: she spends $\frac{3}{5}$ of her wealth on good 1, so $x_1 = \frac{3}{5} \cdot 25/3 = 5$. (Tangency: $\mathrm{MRS}_{12} = \frac{3x_2}{2x_1} = \frac{3}{4}$ and the budget line.)

B4 [3 points] A consumer's indirect utility function is $v(p,w) = \dfrac{w}{p_1 + 3p_2}$. What is her expenditure $e(p,u)$ at prices $p = (2,1)$ and utility level $u = 6$?

Answer: $e = 30$

Working. Duality: $v(p, e(p,u)) = u$ gives $e(p,u) = u(p_1 + 3p_2) = 6 \cdot 5 = 30$.

B5 [4 points] A consumer's expenditure function is $e(p,u) = 2u\,p_1^{3/4}p_2^{1/4}$. The price of good 1 rises from $p_1^0 = 1$ to $p_1^1 = 16$; $p_2 = 1$ stays (useful: $16^{1/4} = 2$). What is her true cost-of-living index $e(p^1,u^0)/e(p^0,u^0)$ (at any utility level u^0)?

Answer: index $= 8$

Working. $e(p^1,u^0)/e(p^0,u^0) = (16/1)^{3/4} = 8$, whatever u^0 is.

B6 [4 points] A consumer has utility $u(x_1, x_2) = \min\{3x_1,\, 4x_2\}$ and wealth $w = 63$. The price of good 2 is $p_2 = 8$; the price of good 1 rises from $p_1^0 = 3$ to $p_1^1 = 6$. What is her compensating variation CV of this price change (negative for a loss, as on page 2)?

Answer: $CV = -21$

Working. Utility u needs $3x_1 \geq u$ and $4x_2 \geq u$, so $e(p, u) = u\,(p_1/3 + p_2/4)$ and $v(p, w) = w/(p_1/3 + p_2/4)$. $u^0 = v(p^0, w) = 63/(1 + 2) = 21$; $CV = w - e(p^1, u^0) = 63 - 21\,(2 + 2) = 63 - 84 = -21$ (negative: a loss).

B7 [4 points] Ivo is an expected utility maximizer with Bernoulli utility $u(x) = \ln x$ over money $x > 0$ in euros. A lottery pays 4 euros or 100 euros, each with probability $\frac{1}{2}$. What is its certainty equivalent, in euros? (You may use $\ln a + \ln b = \ln(ab)$.)

Answer: certainty equivalent $= 20$

Working. $E[u] = \frac{1}{2}\ln 4 + \frac{1}{2}\ln 100 = \frac{1}{2}\ln(4 \cdot 100) = \frac{1}{2}\ln 400 = \ln\sqrt{400} = \ln 20$. The certainty equivalent solves $\ln c = E[u] = \ln 20$, so $c = 20$ euros.

B8 [3 points] Ivo (Bernoulli utility $u(x) = \ln x$) faces another lottery: 1 euro or 49 euros, each with probability $\frac{1}{2}$. What is its risk premium RP (expected value minus certainty equivalent), in euros?

Answer: $RP = 18$

Working. $E[x] = \frac{1}{2}(1 + 49) = 25$; $E[u] = \frac{1}{2}\ln 1 + \frac{1}{2}\ln 49 = \ln 7$, so $c = 7$ and $RP = 25 - 7 = 18$ euros.

Part C (show your steps, 30 points)

Problem C1 [16 points] $u(x_1, x_2) = x_1 + 6\sqrt{x_2}$ on $\mathbb{R}^2_+$.

(a) [5 points] Find the Walrasian demand $x(p, w) = (x_1(p, w), x_2(p, w))$ for $w > 9p_1^2/p_2$.

Steps. For $w > 9p_1^2/p_2$ the optimum is interior. $\mathrm{MRS}_{12} = \dfrac{1}{3/\sqrt{x_2}} = \dfrac{\sqrt{x_2}}{3} = \dfrac{p_1}{p_2}$ gives $x_2 = 9p_1^2/p_2^2$; the budget line gives $x_1 = (w - p_2x_2)/p_1 = w/p_1 - 9p_1/p_2 > 0$.

Answer boxes: $x_1(p, w) = \dfrac{w}{p_1} - \dfrac{9p_1}{p_2}$ [2] $\qquad x_2(p, w) = \dfrac{9p_1^2}{p_2^2}$ [3]

(b) [3 points] Find the indirect utility function $v(p, w)$ for $w > 9p_1^2/p_2$.

Steps. $v = x_1 + 6\sqrt{x_2} = \dfrac{w}{p_1} - \dfrac{9p_1}{p_2} + \dfrac{18p_1}{p_2} = \dfrac{w}{p_1} + \dfrac{9p_1}{p_2}$.

Answer boxes: $v(p, w) = \dfrac{w}{p_1} + \dfrac{9p_1}{p_2}$ [3]

(c) [4 points] Find the expenditure function $e(p, u)$ for $u > 18p_1/p_2$, and the Hicksian demands $h_1(p, u)$ and $h_2(p, u)$.

Steps. Solve $v(p, e) = u$, that is $e/p_1 + 9p_1/p_2 = u$: $e(p, u) = p_1u - 9p_1^2/p_2$ (valid for $u > 18p_1/p_2$, where $h_1 > 0$). Shephard: $h_1 = \partial e/\partial p_1 = u - 18p_1/p_2$ and $h_2 = \partial e/\partial p_2 = 9p_1^2/p_2^2$.

Answer boxes: $e(p, u) = p_1u - \dfrac{9p_1^2}{p_2}$ [2] $\qquad h_1(p, u) = u - \dfrac{18p_1}{p_2}$ [1] $\qquad h_2(p, u) = \dfrac{9p_1^2}{p_2^2}$ [1]

(d) [4 points] Let $p_1 = 1$ and $w = 30$, and let the price of good 2 rise from $p_2^0 = 1$ to $p_2^1 = 3$. Compute her compensating variation CV and her equivalent variation EV of this price change.

Steps. $u^0 = v(1, 1, 30) = 39$, $u^1 = v(1, 3, 30) = 33$ (both interior: $30 > 9$). $CV = w - e(p^1, u^0) = 30 - (39 - 3) = -6$. $EV = e(p^0, u^1) - w = (33 - 9) - 30 = -6$. They coincide: quasilinear preferences in good 1 have no wealth effect on good 2.

Answer boxes: $CV = -6$ [2] $\qquad EV = -6$ [2]

Problem C2 [14 points] $u(x_1, x_2) = \max\{x_1,\, 2x_2\}$ on $\mathbb{R}^2_+$.

(a) [3 points] Are these preferences monotone? Are they strongly monotone? Prove each answer, or give two bundles that show it.

Answer: Monotone yes, strongly monotone no.

Reason. Monotone: if $y \gg x$ then $y_1 > x_1$ and $2y_2 > 2x_2$, so $\max\{y_1, 2y_2\} > \max\{x_1, 2x_2\}$. Not strongly monotone: $(2, \frac{1}{2}) \geq (2, 0)$, both give $u = 2$.

(b) [4 points] Are these preferences homothetic? Are they continuous? Prove each answer.

Answer: Homothetic yes, continuous yes.

Reason. Homothetic: for $\alpha > 0$, $u(\alpha x) = \max\{\alpha x_1, 2\alpha x_2\} = \alpha\, u(x)$, so $u(x) = u(y)$ implies $u(\alpha x) = u(\alpha y)$: $x \sim y \Rightarrow \alpha x \sim \alpha y$. Continuous: u is the maximum of two continuous functions, hence continuous, so the upper and lower contour sets $\{y : u(y) \geq u(x)\}$ and $\{y : u(y) \leq u(x)\}$ are closed.

(c) [3 points] Are these preferences convex? Circle Yes or No. Explain, or give bundles that show it.

Answer: No.

Reason. $(2, 0) \sim (0, 1)$ (both $u = 2$), but the midpoint $(1, \frac{1}{2})$ has $u = 1 < 2$: the upper contour set of $(2, 0)$ is not convex.

(d) [4 points] Is there any concave utility function that represents these preferences? Circle Yes or No and explain.

Answer: No.

Reason. A concave function is quasiconcave, and a quasiconcave representation means convex preferences; (c) shows the preferences are not convex. So no representation, concave or merely quasiconcave, exists.